

Trinh Phan-Canh, MPharm. PhD.

Postdoctoral Researcher — BIDMC, Harvard Medical School

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Education

- Jan 2021 – Oct 2025 **Ph.D. in Molecular Genetics & Immunology**
Medical University of Vienna – Max Perutz Labs Vienna, Vienna Biocenter Campus
 - Thesis: *Antifungal Multidrug Resistance and Virulence of Candida auris* (Grade: 1 – Excellence)
 - Advisor: Prof. Dr. Karl Kuchler
- Sep 2017 – Sep 2019 **M.Sc. in Pharmacy (Highest Distinction – Rector's Award)**
School of Pharmacy, University of Medicine and Pharmacy at Ho Chi Minh City (UMP)
 - Thesis: *Morphology, chemical composition and antimicrobial activity of Dirinaria applanata* (Grade: 10/10)
- Oct 2015 – Feb 2018 **B.Sc. in Information Technology (Rector's Award for Academic Excellence)**
Vietnam National University – Ho Chi Minh City (VNU-HCM)
 - Focus: Management Information Systems (Distance Education Program)
- 2010 – 2015 **National Pharmacist Degree (Rector's Award for Academic & Extracurricular Excellence)**
School of Pharmacy, University of Medicine and Pharmacy at Ho Chi Minh City (UMP)
 - Thesis: *Fermentation conditions for Bacillus subtilis KP3 spores' production* (Rank 1; Grade: 10/10)

Awards and Distinctions

- Life Sciences PHD Awards Austria 2026 — Austrian Society for Molecular Biosciences and Biotechnology (2026)
- ÖGMM Publication Award 2026 — Austrian Society for Medical Mycology (2026)
- Österreichische Mikrobiologie-Preis 2026 (Austrian Microbiology Prize 2026) — ÖGHMP (2026)
- The *Pathogens* Young Investigator Award — Pathogens Journal (2026)
- The *Biology* Young Investigator Award — Biology Journal (2026)
- Bernd Rode Award — ASEA-UNINET Junior Researcher Award (2024)
- Youth Travel Award — FEBS Advanced Lecture Course, France (2024)
- ESCMID Research Grant — Outstanding project in fungal pathogenesis (2023)
- FEMS Travel Grant — Candida and Candidiasis Meeting, Canada (2023)
- Ernst Mach Grant — PhD Fellowship, ASEA-UNINET (2021–2023)
- Pilot International Research Grant — TRAC Sweden-Vietnam (2019)
- Rector's Award, Master's Degree of Pharmacy — UMP Vietnam (2019)
- Youth Leaders Award — Rector's Board, UMP Vietnam (2017–2019)
- Homtamin Scholarship — Korea United Pharm, awarded for academic excellence (2015)
- Daewoong Scholarship — Daewoong Pharmaceutical, awarded for academic excellence (2013)
- First Prize, Scientific Research Contest: Seeking Elixir — UMP (2012)
- Student of Five Merits awards — UMP (2011–2015)
- Honorable Mention — Thua Thien Hue Biology Olympiad (2010)
- Thua Thien Hue Innovation Contest for Youth — awarded for developing a biology learning app (2009)

Featured Publications

The carbonic anhydrase Nce103 controls fungal fitness and skin tropism of Candida auris

Phan-Canh T, Penninger P, Seiser S, et al., Elbe-Bürger A & Kuchler K.

Nature Microbiology, 2026

- This research resolves a 15-year-old question on intrinsic amphotericin B resistance.
- Cover image of Issue February 2026
- Spotlight in Trends in Microbiology (doi.org/10.1016/j.tim.2026.02.009)

White-Brown switching controls phenotypic plasticity and virulence of Candida auris

Phan-Canh T, Jenull S, Bitencourt T, et al. & Kuchler K.

Cell Reports, 2025

- The first comprehensive molecular basis of epigenetic switching in *Candida auris*.



Publications

1. **Phan-Canh T**, Nguyen MLD, Phuc-Loi L, et al. & Kuchler K.
Rapid in vitro evolution of flucytosine resistance in Candida auris
mSphere, 2025
2. **Phan-Canh T**, Bitencourt T & Kuchler K.
Gene dosage of PDR16 modulates azole susceptibility in Candida auris
Microbiology Spectrum, 2025
3. **Phan-Canh T** & Kuchler K.
Do morphogenetic switching and intraspecies variation enhance virulence of Candida auris?
PLOS Pathogens, 2024
4. **Phan-Canh T**, Lackner M, Chauhan M, Zenz L-M, Chauhan N & Kuchler K.
Cytochrome bc1 catalytic subunit Rip1 modulates antifungal tolerance in Candida auris
ACS Infectious Diseases, 2026
5. **Phan-Canh T**, Elbe-Bürger A & Kuchler K.
Antifungal drug resistance in Candida auris: molecular mechanisms and emerging therapeutic options
Trends in Microbiology, 2026 · *Under review*
6. **Phan-Canh T**, Seiser S, Chauhan M, Kuchler K, Chauhan N & Elbe-Bürger A.
Pathogenicity and virulence of Candida auris
Virulence, 2026
7. **Phan-Canh T**, et al. & Chauhan N & Kuchler K.
Antifungal resistance can emerge in Candida auris without drug selection pressure
Microbiology Spectrum, 2026 · *Under revision*
8. Seiser S, Brezovec H, Penninger P, **Phan-Canh T**, et al., Kuchler K & Elbe-Bürger A.
Human keratinocytes and fibroblasts coordinate early cutaneous innate defenses against Candida auris
Journal of Investigative Dermatology, 2026
9. Tsymala I, Penninger P, Nogueira F, Jenull S, **Phan-Canh T**, et al. & Kuchler K.
Innate lymphoid cells integrate sensing and plasticity to control fungal infections
Cell Reports, 2026
10. Bitencourt T, Tsymala I, et al., **Phan-Canh T**, et al. & Kuchler K.
Candida-Klebsiella interactions rewire fungal morphogenesis to create a host environment favoring pathogen survival with enhanced tissue pathology.
bioRxiv (preprint), 2026
11. Cerbu D, Seiser S, **Phan-Canh T**, et al., Kuchler K & Elbe-Bürger A.
Octenidine effectively reduces Candida auris colonisation on human skin
Scientific Reports, 2025
12. Kavaklioğlu G, Podhornik A, Vcelkova T, Marjanovic M, Beck MA, **Phan-Canh T**, et al.
L1TD1 associates with ORF1p to promote LINE-1 retrotransposition
eLife, 2025
13. Penninger P, Brezovec H, Tsymala I, Teufel M, **Phan-Canh T**, et al. & Kuchler K.
HDAC1 fine-tunes Th17 polarization in fungal infections

14. Bitencourt T, Nogueira F, Jenull S, **Phan-Canh T**, et al., Kuchler K & Lion T.
Multiomics reveals interspecies interaction pathways between *A. fumigatus* and *K. pneumoniae*
Communications Biology, 2024

15. Seiser S, Arzani H, Ayub T, **Phan-Canh T**, et al. & Kuchler K & Elbe-Bürger A.
Human and mouse skin models to study *Candida auris* infection
Microbes and Infection, 2023

16. Jenull S, Shivarathri R, Tsymala I, Penninger P, **Phan-Canh T**, et al. & Kuchler K.
Transcriptomics defines echinocandin resistance in *Candida auris*
mBio, 2022

17. Shivarathri R, Chauhan M, Mazumdar R, **Phan-Canh T**, et al. & Kuchler K & Chauhan N.
A proteomic approach for the quantification of post-translational protein lysine acetylation in *Candida albicans*
Methods in Molecular Biology (Candida species), 2022



Outreach Publications

1. Phan-Canh T, Elbe-Bürger A & Kuchler K.
A Breath to Persist. CO₂ Fixation in *Candida auris*
SPECTRUM Dermatologie, 2026

2. Phan-Canh T.
Scientific mobility in microbiology – 19
Trends in Microbiology, 2026



Abstracts, Posters, and Presentations

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Mar 2026

Talk — ÖGMM Symposium 2026, im Rahmen des ÖIK, Saalfelden, Austria

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Jan 2026

Talk — Mycology Meeting, Institute of Hygiene and Medical Microbiology, Innsbruck, Austria

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Oct 2025

Poster & Elevator Talk — *Candida and Candidiasis*, Microbiology Society, Berlin, Germany

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May 2025

Poster & Poster Slam — Annual YSA PhD Symposium, Medical University of Vienna

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May 2024

Poster — *Host–pathogen interactions and virulence in human fungal pathogens*, FEBS Advanced Lecture Course, France

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Mar 2023

Poster — *Candida and Candidiasis*, Microbiology Society, Montreal, Canada

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Jan 2023

Oral — *Immune Control of Tissue Homeostasis*, TissueHome Symposium, Vienna, Austria

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Nov 2022

Poster — VBC PhD Symposium, Vienna Biocenter Campus

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Sep 2022

Oral — *Scientific Seminar of Young Vietnamese Intellectuals*, Prague, Czech Republic

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May 2022

Poster — *Host–pathogen interactions and virulence in human fungal pathogens*, FEBS Advanced Lecture Course, France



Professional Experience

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06/2026 – present

Postdoctoral Researcher
BIDMC, Harvard Medical School

- Development of spatial multi-omics technologies for host–disease interactions

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1–5/2026

Short-term Visiting Postdoc
Institute of Hygiene and Medical Microbiology, Innsbruck, Austria

- Trained PhD students and staff in genetic engineering and bioinformatics
- Led a research project in antifungal drug discovery

- 2021 – 2025 **PhD Candidate, Karl Kuchler Lab**
Max Perutz Labs, Vienna Biocenter
 - CRISPR-based genetic manipulation, multi-omics analysis (WGS, RNA-seq, proteomics, lipidomics)
 - Host-pathogen and interkingdom interactions (yeast, in vitro with immune cells, in vivo mouse models)
 - Drug resistance, microbial immunology and fungal pathogenesis
- 2016 – 2021 **Lecturer**
School of Pharmacy, UMP, Vietnam
 - Led antimicrobial discovery projects using Myxobacteria and Streptomyces
 - Supervised undergraduates and coordinated the Mycology Laboratory
- 2015 – 2017 **Pharmacist & System Administrator, Drug Interactions Checker**
thongtinhuoc.com.vn
 - Built and maintained drug info database; led safety content review and UX
- 2013 – 2015 **Undergraduate Research Assistant**
Pharmaceutical biotechnology lab, UMP, Vietnam
 - Investigation of sporulation dynamics in Bacillus subtilis during large-scale production
 - Mathematical models to optimize sporulation production of B. subtilis

Other Technical Skills

- Wet lab techniques:** Cloning, flow cytometry, biochemical assays, mouse models, microscopy
- Programming & Data Analysis:** Proficient in R; working knowledge of Python, Bash, SQL
- Omics Analysis:** WGS, RNA-seq, proteomics, lipidomics (design, processing, interpretation); basic knowledge of scRNA-seq and spatial transcriptomics
- Visualization:** ggplot2, GraphPad Prism, FlowJo, Adobe Illustrator, Photoshop, InDesign
- Languages:** English (professional fluency), Vietnamese (native)

Reviewing Experience

- Editor-in-Training**, International Journal of Antimicrobial Agents
- Regular **Reviewer** for ASM Microbiology Spectrum, Pathogens and Disease, Fungal Genetics and Biology

Mentoring and Teaching

Supervision and mentorship of students (Master's, Diploma, Internships). Successfully supervised multiple students whose research projects received national awards:

- Best Diploma Thesis Award (High School Level), Austria — 2023**
Mentee: Maja Prettnner and Katja Walter
- National Eureka Award research competition, Vietnam — 2019**
Mentee: Duong Nguyen Phi Long
- Finding Elixir Competition at UMP, Vietnam — 2018**
Mentees: Tran Quoc Vuong, Nguyen Xuan Loc, Nguyen Thi Loan
- Summer Research Program at UMP, Vietnam — 2017, 2018, 2019**
Mentees: Vo Linh Tu, Pham Tran Kim Quyen, Tran Thi Thu Ha
- Developed curriculum and led student research groups at UMP — 2016–2021**

Leadership and Outreach

- Founder, PhD Talk: Seminar series for Vietnamese biologists in Austria — phdtalk.github.io (2021–2022)
- President, Vietnamese Student Association in Austria (2022–2024)
- Organizer of national and university science conferences (2020–2021)
- Member, Executive Committee, Vietnamese Youth and Student Federation in Europe (2021–2024)

- Editor, Pharmacy Brief News: A bilingual outreach publication (2020)
- Founder, President, Pharmacy Soft Skill Club (2011–2015)

References

Karl Kuchler, PhD
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